MILITARY SPECIFICATION

TUBES, STOMACH, SURGICAL, PLASTIC, RADIOPAQUE, SALEM, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers sterile, disposable, radiopaque, double-lumen stomach tubes of the Salem design, suitable for use in aspirating stomach contents.

1.2 Classification. The stomach tubes shall be of the following sizes as specified (see 6.1):

Sizes	French
1	10
2	12
3	14
4	16
5	18

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water Resistant, (For Sealing Fiberboard Boxes).
PPP-B-566	Box, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-plywood.
PPP-B-621	Box, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-up.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water-Resistant, (For Carton Sealing).

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Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-vaporproof or Waterproof, Flexible.
MIL-S-36586	Sterilization Strip Test; Bacterial Spore.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

24253	TUBES, STOMACH, SURGICAL, PLASTIC, Radiopaque, Salem, Disposable.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pannsylvania 18042).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-412	Tension Testing of Vulcanized Rubber.

(Application for copies should be addressed to American Society for Testing and Materials, 1903 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Tubing. The tubing shall be of a medical grade, clear, transparent, polyvinyl chloride plastic. The tube shall meet United States Pharmacopeia (USP) requirements for Class IV Plastics, and for Transfusion and Infusion Assemblies.

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3.2 Style, design, and dimensions.

3.2.1 The sterile, disposable, plastic stomach tube shall have a radiopaque line running lengthwise within its wall. It shall be closed at the distal end and have ten oval eyes also at the distal end. The proximal end shall be flared to accommodate a removable connector. Also within the wall of the main lumen, there shall be a smaller secondary lumen which opens to the main lumen at the distal end. The secondary lumen serves as an air vent and can be used for irrigation also. At the proximal end of the secondary lumen a "cap" integral with a short piece of tubing shall be permanently attached. The "cap" shall fit hermetically over the connector. The tube shall be marked to show depths of insertion. The dimensions and markings shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 24353.

3.2.2 Radiopacity. The tube shall have a radiopaque stripe of uniform width for its entire length.

3.3 Physical properties. Breaking strengths and percentage elongations for the stomach tubes shall be as listed in Table I when tested as specified in 4.4.4.

Table I. Physical Properties

Size	Breaking strength (pounds min.)	% elongation (min.)
1	12	175
2	18	175
3	24	175
4	29	175
5	36	175

3.3.1 Hardness. The Shore Durometer Hardness, Class A, shall be 77 $\pm$ 3 when tested according to 4.4.7.

3.4 Sterility. The stomach tubes shall be sterile when tested with a biological indicator as specified in 4.4.1.

3.4.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

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a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.4.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.5 Safety (non-toxicity) and non-pyrogenicity. The stomach tubes shall be safe and pyrogen-free when tested as specified in 4.4.2 and 4.4.3.

3.6 Radiopacity. The stomach tubes shall have a radiopaque stripe that shows as a definite, uniform shadow when tested as specified in 4.4.5.

3.7 Certificates of sterility, freedom from pyrogens, safety, biological tests and animal implantation tests. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested and found to be sterile, free from pyrogens, nontoxic (safe), and have passed the biological tests and the animal implantation test. This certificate shall include the specific tests performed and the tests results. The record of the animal implantation test and its results shall be certified by the pathologist who performed or supervised performance of the animal implantation test. For destination

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inspection, certificates shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical Materiel, Technical Operations Division, 2800 South 20th Street, Philadelphia, Pa. 19101, attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

3.8 Resistance to autoclaving. The stomach tubes shall withstand autoclaving, as specified in 4.4.8, without visible deterioration other than some whitening of the stomach tubes.

3.9 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label, which can be verified against the manufacturer's records of all tests made of animal implantation, sterility, and non-pyrogenicity. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.10 Workmanship. Workmanship shall be first class throughout. The finished stomach tubes shall be symmetrical in shape, and the surface shall have a smooth finish, free from irregularities or imperfections of any kind which detract from their appearance or which may impair their serviceability. The stomach tubes may be visually clean and free from foreign matter and soils.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105 and as indicated in Table II. The unit of product for examination shall be one stomach tube.

Table II. Sampling for Examination

Type defect	Inspection level	AQL (percent defective)*
Major	S-2	1.0
Minor	S-2	1.5

* If acceptance number is not exceeded, lot shall be accepted. If rejection number is equaled or exceeded, inspection at level II shall be performed to determine final rejection or acceptance.

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and as shown in Table III.

Table III. Sampling Criteria for Tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent) defective)
Class IV Plastic	3.1 & USP	4.4.6 & USP	S-2	*
Breaking strength	3.3	4.4.4	S-2	2.5
Elongation	3.3	4.4.4	S-2	2.5
Hardness	3.3.1	4.4.7	S-2	4.0
Sterility	3.4	4.4.1	S-1	*
Safety (non-toxicity)	3.5	4.4.2	S-1	*
Non-pyrogenicity	3.5	4.4.3	S-1	*
Radiopacity	3.6	4.4.5	S-1	*
Autoclaving	3.8	4.4.8	S-1	*

* Acceptance number shall be zero. If rejection number is equaled or exceeded, inspection at level II shall be performed to determine final rejection or acceptance.

4.2.3 Test samples. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

4.3 Examination. The stomach tubes shall be examined to determine compliance with all requirements contained in the specification. Examination shall be conducted as specified in Tables II and IV. Examination, however, shall not be restricted to defects listed in Table IV. Any stomach tubes in the sample, containing one or more defects shall be rejected (defective) and if the number of defective stomach tubes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected (see 4.2.1).

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Table IV. Classification of Defects

Categories	Defects
Major	
101	Rough edges to eyes or distal end.
102	Foreign matter in or on stomach tube.
103	Presence of embedded particles.
104	Tackiness.
105	Dimensions out of tolerance.
106	Surface not smooth.
107	Connector not present.
Minor	
201	Eyes substantially not centered relative to radiopaque line.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Sterility test. The stomach tubes shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybeam casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.1.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table III, Sampling Criteria for Tests.

4.4.1.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

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4.4.1.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.1 Sterility test and 4.4.1.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.1.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.1.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.1.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.4.2 Safety. Tubes shall be tested for safety in accordance with the USP for Transfusion and Infusion assemblies.

4.4.3 Non-Pyrogenicity. Tubes shall be tested for freedom from pyrogens in accordance with the USP for Transfusion and Infusion assemblies.

4.4.4 Breaking strength and elongation. The stomach tubes shall be tested for breaking strength and ultimate elongation in accordance with ASTM D-412, Tension Testing of Vulcanized Rubber.

4.4.5 Radiopacity. Sample stomach tubes shall be placed on an unexposed X-ray plate and covered with a 5/8 inch thickness of aluminum, X-ray exposure shall be 50 milliampere-seconds (MAS), 82 kilovolt peak (KVP) at 30 inches using M-speed screens in a convenient size cassette.

4.4.6 Animal implantation test. Samples of the stomach tubes shall be tested in accordance with method given in the USP for Implantation Test, except for the changes which follow: Last paragraph, second line after "72 hours", add "and not to exceed 144 hours." Last paragraph, third and fourth sentence, delete entirely and substitute "The tissues surrounding the implants shall be examined, both grossly and microscopically, by a pathologist certified by the American Board of Pathology or the American College of Veterinary Pathologists."

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5.1.2.2 Unit package. Twenty five size 1 individually packaged tubes or fifty individually packaged tubes of one size, sizes 2 thru 5, shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, type CF, class weather-resistant. Closure shall be effected by taping the top and bottom flaps with two inch wide tape conforming to PPP-T-76. Closure of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Four unit packages of size 1 tubes or six unit packages, of one size tubes only, sizes 2 thru 5 shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. The box shall conform to the special requirements as specified in PPP-B-636. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. Four unit packages of size 1 tubes or six unit packages of one size tubes only, sizes 2 thru 5, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. The box shall conform to the special requirements as specified in PPP-B-636. Closure of the box shall conform to method I or II as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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